

MADHVIKA SEHGAL

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PROFESSIONAL SUMMARY

Business Intelligence Engineer with 4+ years of experience at Amazon and Barclays. I build SQL-based ETL pipelines and data models, then use them for analytics, reporting, and dashboards across Finance, Operations, Product, and Customer Care. My work is mainly in SQL, Python, and AWS, with recent use of Claude-assisted workflows for debugging, documentation, and reporting automation.

TECHNICAL SKILLS

Languages & Query: SQL, Python (Pandas, NumPy, Jupyter), Spark SQL, SAS, Advanced Excel

Data Engineering: ETL/ELT, Data Warehousing & Dimensional Modeling, Data Quality & Reconciliation, Schema Migrations, Git

Cloud & BI Platforms: AWS (Redshift, Athena, S3, Glue, Lambda), Oracle, Tableau & Tableau Prep, Amazon QuickSight, Alteryx

AI & Automation: Amazon Bedrock (Claude) for LLM-assisted SQL investigation and reporting automation

PROFESSIONAL EXPERIENCE

Business Intelligence Engineer | **Amazon – Health Finance & Analytics** | Seattle, WA | Sep 2025 – Aug 2026

- **Cost Allocation:** Designed a 1,500+ line **Spark SQL cost-allocation model** across 7+ operating cost pools, replacing payroll-based site proxies with productivity-hour drivers and reconciling finance and ops data at different grains for more accurate allocation.
- **Small Format Economics:** Re-architected the Small Format and Kiosk cost-attribution framework that separated blended labor costs, **improved warehouse-level unit economics**, and informed the business case to **scale the network from 4 to 38 sites**.
- **Platform Governance:** Built a governance and adoption **dashboard** from **1M+ monthly audit events**, flagging underused and high-compute jobs; drove retirement of 14 stale jobs and saved roughly 1 hour/day in compute time.
- **Workforce Cost Automation:** Automated monthly workforce-cost allocation in SQL using **ADP payroll and warehouse metadata**, replacing a manual Excel process and saving **7+ hours per month**.
- **Schema Migration:** Led downstream remediation for a **primary-key and schema migration** across 10+ datasets, adding validation guardrails to ensure **zero metric regression**.
- **AI-Assisted Workflow:** Built a **Claude-assisted workflow** to investigate month-end GL anomalies, generating candidate root causes and validation SQL for analyst review; cut debugging time by **about 3 hours per issue**.

Business Intelligence Analyst | **Barclays – Credit Card, Contact Center Analytics** | Whippany, NJ | Jan 2021 – Jul 2025

- **Callback Feature Revamp:** Led a 3-member team to redesign SLA logic and integrate callback features into Contact Center workflows. Updated 45+ dashboards and reports, contributing to **~12% higher customer retention and ~8% fewer transfers**.
- **ETL Framework for Call Analytics:** Built and managed end-to-end ELT pipelines and curated IVR data models powering **30+ Tableau dashboards**, standardizing call, routing, containment, and funnel metrics across reporting.
- **Agent Performance Scoring Model:** Developed a scoring model using sentiment, CSAT, and NPS data to benchmark agent performance and identify outliers for QA and coaching.
- **Customer Journey Analytics:** Analyzed clickstream, call-log, and servicing data across **5M+ interactions**. Built Tableau heatmaps and journey analysis that helped reduce **average call time by ~15% and transfers by ~20%**.
- **Real-Time Marketing Insights:** Developed Redshift-powered dashboards for real-time KPI tracking, contributing to a **~20% increase in containment and ~80% reduction in manual reporting effort**.
- **Funnel Monitoring:** Built dashboards for containment, self-service, sentiment, and funnel KPIs; introduced tagging improvements that reduced transfers by **~30%**.
- **GenAI-Driven Automation:** Engineered a serverless solution using **AWS Lambda and Claude via Amazon Bedrock** to automate report metadata summarization, reducing audit and documentation work from **days to minutes**.

Business Consultant | **Accenture – SAP MM** | Delhi, India | Jan 2017 – May 2019

- **Led end-to-end SAP MM delivery** for global clients — from requirements gathering and BRDs, through system enhancements built with ABAP developers, to UAT and deployment — supporting **95% user adoption**.

EDUCATION

- **Master of Science, Management Information Systems**

University of Arizona, Tucson, AZ | Aug 2019 – Dec 2020

- **Bachelor of Technology, Information Technology**

Guru Gobind Singh Indraprastha University, New Delhi, India | Aug 2012 – May 2016